

Differentiated Compensation Systems to Improve Teacher Effectiveness and Retention

Introduction

Though education in the United States aims to accomplish a variety of goals, one objective that most can agree on is that students are meant to learn and grow in school, and this is a direct consequence of how well they are instructed by their teachers. Since Ronald Reagan's *A Nation at Risk* report in 1983 and George W. Bush's No Child Left Behind Act, US schools have shifted to a stricter focus on accountability and ensuring students meet performance criteria, as measured by standardized tests (Springer, 2023, p.69). Teachers are at the heart of achieving this goal, as they have the most potential to impact student outcomes, so it makes sense to attempt to intervene at the teacher-level to target student achievement. Research has proven that teachers are the most impactful within-school determinant of growth. Rivkin, Hanushek et al. (2015) argue that even within the same school, achievement varies greatly due to teacher differences, and higher teacher quality is even more beneficial than reducing class size (p.417). Effective instruction can help low income students overcome the achievement gap as well. As a concrete example, it was estimated that having a teacher with effectiveness one standard deviation above the mean is associated with an added value of \$500,000 to future student earnings (Springer et al., 2016, p.199). Therefore, attracting, retaining, and building up more effective teachers has the potential to significantly improve student achievement and reduce disparities.

Scholars continue to question how policies might lead to increased effectiveness of educators. One option is reforming compensation structures. Traditional uniform pay scales do not reward nor reflect teacher effectiveness. Most current salary systems use years of teaching experience and highest degree earned as the two most important factors in determining salaries (Springer, 2023). These are not the strongest predictors of teacher effectiveness and student achievement. An abundance of literature provides evidence that master's degrees do not have a

significant impact on student achievement (Rivkin, Hanushek et al., 2015, p.419). Hence, some researchers argue that the single-salary schedule– in which teachers with the same basic qualifications receive the same pay– does not necessarily align with the major goals of education, including helping students meet learning metrics. In other professions, there is greater pay flexibility depending on market conditions, and in higher education, pay depends on specific fields (Springer, 2023). These methods might be applied to K-12 schooling as well. Springer also argues that single-salary schedules disadvantage traditionally marginalized students.

Consequently, differential compensation for teachers might combat the inefficiencies seen in uniform pay schedules, leading to better outcomes for students. Under the Teacher Incentive Fund, the US Department of Education has awarded grants to schools to implement performance-based compensation programs (Springer, 2023). The idea is that the funds will improve teacher effectiveness to drive positive student outcomes, especially in high-needs schools. From 2006 to 2010 around \$700 million were allocated to performance pay programs (Springer, 2023, p. 69). Later on, Obama’s Race to the Top (RTTT) grants helped fund similar systems, so there are plenty of differential pay programs to evaluate.

In this analysis, I aim to answer the question of how compensation plans can be restructured to have the most positive impacts on student learning. I examine two different options for reforming the typical single-salary schedule: paying teachers more based on test score performance versus allocating bonuses aimed at attracting and retaining high quality teachers, particularly in hard-to-staff schools. In both cases, the main goal is to enhance student achievement as much as possible through effective teaching. Improving access to quality education is crucial since this plays a large role in student growth. Policies providing incentives to teachers can have different approaches– either motivating better teaching strategies to help

students learn (typically as measured by test scores) or convincing them to remain in the same school for longer periods of time, which also has been shown to enhance learning. Scholars believe that more differentiated pay methods including merit-based bonuses will motivate teachers to both align their practices to better help students meet performance metrics and remain in their schools for longer (Pham et al., 2020). Opponents present several counter arguments. For instance, some argue that it is impossible to fairly evaluate teacher performance to effectively reward those that supposedly “deserve” it. Many efforts at financial incentives for teachers have been unsuccessful, but this might be attributed to ill-intentions and loosely carried out plans. Some districts implemented the program simply for the sake of appearing progressive, which failed, but other districts have seen success at improving retention and meeting learning goals (Springer, 2023, p.70). Comparing merit pay policies aimed at increasing test scores to bonuses designed to attract and retain teachers, I conclude that there is the most potential for attraction and retention bonuses to succeed at making a difference in student achievement, but programs must be designed well to satisfy all stakeholders.

Option 1: Merit Pay for Test Score Improvements

Some school districts have opted to incentivize teachers through increased pay if they perform well. The motivating theory behind these teacher-level performance incentives comes from the economics principal-agent theory. When compensation methods are not based on metrics that align with school goals to improve learning, then teacher efforts will not be directed toward the most efficient places. If teachers are compensated according to how well they promote student growth, then they will be more motivated to focus their efforts on meeting that goal (Pham et al., 2020). This still leaves room for teachers to make autonomous decisions about

how they want to run their classrooms, so they are not overly restricted by the merit pay policy. The main measurement to assess merit pay programs has been test scores. In a meta-analysis synthesizing the results of 26 merit pay programs in the United States, Pham et al. found an overall positive impact of merit pay programs on test scores. On average, the effect of teacher participation in a US merit-pay program was associated with a 0.043 standard deviation increase in student test scores (p. 543). To put it into perspective, this effect size is about the same as students receiving an additional 3 weeks of learning. However, impacts vary greatly depending on program design and implementation. There is huge variation in the types of merit pay programs that have been tested. Some reward individual teacher performance while others incorporate school-level incentives. There are also variations in the size of the grants and the durations of programs.

Honing in on one particular example, in Austin, the REACH program was a multifaceted incentive aimed at recruiting, retaining, and rewarding highly effective teachers. Balch and Springer (2015) analyzed the impacts of REACH in depth. One component of the program was awarding teachers based on student growth accomplishments. Teachers were assessed based on student learning objectives and value added to test scores. The SLOs were constructed by teachers themselves (then approved by principals and REACH staff), so they had flexibility to determine what their students needed most. The compensation for meeting SLOs was up to \$1500 for each goal met (p. 116). After the first year of the program, reading and math test scores improved significantly, and this gain persisted into the second year, but beyond that there was no further increase. The positive results from the first year suggest there is potential for new compensation models to make teachers more effective. It is unclear why the effects did not last, though, so overall, with only one year of benefits, the REACH program was not a huge success.

Another example to examine is the Project on Incentives in Teaching (POINT). This Nashville program began in 2006, and the plan was to offer rewards of up to \$15,000 to middle school math teachers if they led students to large gains on tests (Springer, 2023, p.71). Teachers were given the flexibility to decide what methods they would use to increase test scores, which could include professional development, coaching from peers, and/or collaboration. The main goal was to test the hypothesis that a financial incentive in and of itself would be enough of an intervention to motivate teachers to augment student achievements. Ultimately, no evidence was found to support the hypothesis. The only exception was that in fifth grade, the bonus appeared to have a positive effect on test scores. A potential reason for this is that compared to other middle school teachers, fifth grade teachers more frequently teach the same class of students in multiple subjects, not just math (Springer et al., 2010). This lines up with Pham et al.'s meta-analysis that demonstrated stronger impacts in elementary schools than middle schools, assuming that elementary school teachers have the ability to more significantly impact student learning since they are in charge of all subjects. The overall conclusion to be made from the POINT program, therefore, is that merit pay—with no guidance for how teachers should improve their effectiveness—is an ineffective compensation methodology.

Though the overall analysis by Pham et al. and some more specific case studies demonstrated increased test scores following merit pay implementation, there are many cons and frustrations associated with the policy. Firstly, the successes of certain implementations of teacher performance incentives cannot be attributed to the independent impact of a financial bonus. Results from the meta-analysis of Pham et al. demonstrated the strongest positive effects in cases where programs incorporated professional development as well, hinting that there may be confounding factors leading to the perceived test score increases. In Austin, the REACH

program consisted of many components, so it is impossible to isolate the effects of just the payments. For example, the program contained a component of professional development, with stipends for completing certifications and mentoring. Murnane and Cohen (1986) also found that a common characteristic of successful, long-lasting merit pay plans was that they all existed in desirable districts with good working conditions and comparatively high base salaries (p.11). Though these plans had high support from teachers, they did not lead to significant changes in the way teachers taught. This implies that the pre-existing conditions of the school might be what led to the longevity of those merit pay programs. The intended effect of motivating teachers to improve their strategies was not achieved.

Another issue with the merit pay policy is that it largely relies on test scores to evaluate teacher performance, which comes with a host of qualms. Teachers in the POINT program were given surveys to assess their attitudes toward the incentive, and more than half felt the program would do a poor job of correctly identifying the effective teachers (Springer et al., 2010, p.37). They thought it was unfair that just test scores were used as measurements. It is near impossible to define a reliable teacher evaluation strategy. Research found that SLOs were not correlated with additional value to test scores, so it cannot be relied on as a valid measure of teachers' abilities to enhance student learning (Balch and Springer, 2015). Additionally, there are a plethora of external factors that predict student outcomes, such as family environments and adverse life events, so it is incorrect to assume that the only input that determines student test scores is teacher quality. Because teachers find it unfair to base pay solely on what their students accomplish, their morale might be dampened, achieving the opposite effect of what the program intends (Leigh, 2012). To further prove the point, according to Balch and Springer (2015), an analysis of merit pay program surveys suggests that teachers receiving merit awards are less

satisfied with their jobs on average (p.115). A dissatisfied teacher will likely be less effective and less motivated to help their students succeed, so attaching accountability and consequential outcomes to test scores does not appear to be a valid strategy.

Alongside the issue of suitable evaluation metrics, attaching teacher salaries to test scores might inspire teachers to narrow their focus too much on teaching to tests. In turn, they would ignore important untested skills, including both academic content mastery, social-emotional skills, and preparedness for democratic citizenship. Teachers might sacrifice time spent on these untested subjects and life skills solely to improve scores and receive their bonus. It also may influence teachers to focus their attention mostly on students at the cusp of reaching proficiency, which is not fair to other students. In this case, even if test scores increase, this does not mean that educational goals are actually being accomplished.

Though the main intent behind performance-based pay is that it will better motivate teachers to align their efforts with educational goals, research has shown that most teachers in these programs do not change their behavior much. Murnane and Cohen (1986) found that merit pay does not actually motivate teachers. When teachers received evaluation scores lower than expected, this defeated their motivation and caused them to be less willing to work hard. Principals also reported that prior to merit pay, they often gave teachers higher ratings to boost confidence, morale, camaraderie, and overall performance (p.9). With merit pay, they felt pressured to limit the number of high ratings in order to meet cutoffs and budget requirements.

A final issue is that merit pay programs assume teachers have all the resources they need to enhance their effectiveness. Pham et al. (2020) cite research suggesting that teachers are highly motivated without any financial incentive, but they simply do not have the capacity to improve upon what they currently do (p. 531). This might be because of a lack of resources and

overall school funding or a lack of skills, knowledge, and guidance to improve. In the POINT study, over 80 percent of participating teachers claimed to feel that they were already working as effectively as possible, even before the bonus incentives (Springer et al., 2010, p.38). The financial incentive did not cause teachers to change their instruction styles much, if at all. Hence, there were no significant results to come out of the experiment. The incentives were not accompanied by professional development or additional training to help teachers understand *how* to enhance their effectiveness. This could explain why financial incentives failed to improve teacher quality in many cases, and a strategy aimed at teacher training and development might be more effective. Alternatively, if schools find that training teachers to enhance effectiveness is too difficult or costly, they might instead aim to retain their highest-quality teachers for as long as possible, which leads us to the next teacher compensation policy option.

Option 2: Bonuses to Attract and Retain Effective Teachers

A second attempt at a non-traditional pay schedule has been introducing bonuses intended to attract and retain teachers. The motivating framework behind this approach is the fact that frequent turnover of teachers has negative impacts on student learning. Most notably, this approach more directly targets reducing inequalities like unequal access to high quality teachers. Not only has research proven that schools with high proportions of minority and low-income students are less likely to have teachers with high effectiveness, but these schools also tend to have higher rates of turnover (Springer et al., 2016, p.199). Nationally, 21% of teachers leave high-poverty public schools on average, compared to 14% of low-poverty school teachers (Fulbeck, 2014, p.69). This is a particularly pressing issue because these are arguably the students that would benefit most from high quality teaching. In addition, Fullbeck mentions how

first year teachers have been proven to be less effective than experienced teachers, so there is even greater motivation for schools to retain their experienced teachers. Springer (2023) argues that traditional uniform salary schedules contribute to the inequitable distribution of high-quality teachers and so differential pay policies may help rebalance the distribution (p.74). His reasoning is that financial incentives allow schools to keep up with market conditions and retain effective teachers in less competitive or desirable schools such as rural schools or chronically low performing ones. Because of this, targeting high-quality teacher attraction and retention is a reasonable strategy to help improve student learning, especially in high-needs schools. It is possible that one of the main reasons teachers leave their jobs is because they are dissatisfied with their pay. It also makes sense that schools in high-poverty regions are less attractive to teachers who might be able to find better salaries and working conditions in other locations or at private schools, for example. As a result, new payment schemes might convince teachers to move to high-needs schools or stay in their schools for longer periods of time, which in theory should benefit student achievements. As opposed to previously mentioned programs that were designed specifically with the intention of improving test scores, some other example programs designed their systems with the main goal of retaining teachers, which shaped the plans in a different way.

Balch and Springer (2015) cite multiple studies that found retention bonuses, particularly in hard-to-staff schools, reduced turnover rates and improved the composition of teacher quality. One positive example can be seen from Tennessee's initiative to provide retention bonuses (Springer et al., 2016). In Tennessee's Priority Schools, identified as those in the bottom 5%, effective teachers were offered \$5000 as a retention bonus (p. 201). When teachers at high percentiles leave a school, they are on average replaced by teachers at the 25th percentile, so

there is a clear motivation to retain those high performing teachers. The bonus program was isolated from any other school reforms, so its results reveal the true impact of the bonus alone, uninfluenced by other confounders. Springer et al. found that overall, there was no effect of the bonuses on teacher retention, but looking within the group of educators that teach tested subjects, there was a significant positive impact. The highly effective teachers eligible for a bonus were 20% more likely to remain in the school compared to teachers just below the cutoff for the bonus (p.210). The stronger retention also led to increased performance on reading and math tests, demonstrating the positive effects of keeping effective teachers in the school. The fact that retention bonuses mainly motivated teachers of tested subjects is a potential weak point of the policy. Also, in this case, only the most highly effective teachers at low-performing schools were given the bonus. It is plausible therefore that we could see even greater effects if the bonus was offered to a larger group of adequately effective teachers (p.217). Springer also argues that bonuses for effective teachers are better than punishments for ineffective teachers because dismissing teachers leads to more turnover and instability that negatively impacts student achievement and presents the financial challenge of hiring new teachers. All things considered, the Tennessee case provides promising evidence that bonuses can help retain teachers, including in high-needs schools where turnover rates are typically high. Combining this with research on benefits of retention, the bonuses likely help high-needs students grow closer to achieving proficient test scores because they maintain access to a high quality teacher.

As another example, in 2006, Denver Public Schools tested out a new method of compensation in an attempt to reduce teacher mobility, believing that substantial movement makes it difficult for schools to retain a high quality set of educators (Fulbeck, 2014, p.67). Fulbeck analyzes the impacts of this Professional Compensation System for Teachers, a.k.a.

“ProComp”. ProComp enabled teachers to receive a range of bonuses on top of their base salaries from \$376 up to over \$3000 for meeting certain incentives, and those bonuses would aggregate if multiple criteria were met (p.68). One of these incentives was a bonus for remaining in high-poverty schools. She ultimately found that the ProComp incentive led to an overall 30% decrease in the odds of a teacher departing (p.74). Additionally, Atteberry and Lacour (2020) found that ProComp led to more effective teachers being recruited into the Denver Public School System (p.1). Further, their analysis showed that ProComp was associated with better retention of high quality teachers compared to their lower quality peers. These effects led to an overall positive transformation of the teacher workforce with the quality of newly attracted teachers being higher than those who left (p.38). However, the decrease in departures was smaller for teachers in high-poverty schools, suggesting the ProComp was not a strong enough incentive to keep teachers at those schools where retention is most important. This suggests that other factors like school administration, environments, and conditions are influential in driving teachers to stay or leave. As another positive example of retention bonus payments, in North Carolina, they offered an additional \$1800 yearly to math, science, and special education teachers which led to a 17% decrease in teachers departing their roles (Fulbeck, 2014, p.69). Many teachers in this program claimed they did not fully understand the criteria for acquiring the bonus, so there was potential for even greater impacts had all eligible teachers been aware.

Beyond efforts largely to retain teachers, there have been compensation programs designed to attract teachers to fill roles in high needs schools or specific subjects with high vacancy rates. For instance, Pham et al. mention the Talent Transfer Initiative (TTI), which offered \$20,000 to high performing teachers to transfer to low-performing schools. This was a success for the 2 years the intervention lasted, but after the one-time payment, retention rates

remained the same (p. 551). In sum, initial studies of both attraction and retention attempts reveal optimistic outcomes, but there are of course some limitations to be aware of.

One difficulty with a retention bonus policy is that it is unclear what dollar amount of bonus is most appropriate and cost-effective. With few examples to learn from, it is difficult to determine a price that is both effective and does not present a burden to districts. Transparency and evaluation logistics have been another key problem. Due to confusion about the ProComp policy and its results, many teachers in Denver were opposed to the new compensation model and they organized a strike in 2019 (Attebury and Lacour, 2020). Even though district leaders saw rising retention rates after implementation of the program, teachers felt that retention was declining in reality. Administrative reports claimed that without ProComp, attrition rates would have been worse, and since DPS is so large, effects varied greatly by school, leading to confusion about results. Additionally, ProComp was complex with its 10 different categories for bonuses, which made it difficult for teachers to estimate their yearly salaries, further contributing to their frustrations (Attebury and Lacour, 2020, p.39-40).

In Tennessee, an evaluation problem that arose was that the retention bonuses were only given to “highly effective” teachers, as measured by test scores and observations from principals. It is hard to say if these evaluations are reliable. Principals could have inflated scores of certain teachers with the observation component, and if teachers are incorrectly categorized, then the conclusions do not hold since that would mean the ineffective teachers are being retained. Additionally, the Tennessee initiative produced no significant results for teachers of untested subjects, which is likely a consequence of the fact that test scores were included in evaluations, so teachers of different subjects were evaluated differently.

The metric of retention rates itself has potential flaws too. Increased teacher retention could have negative consequences if lower-performing teachers choose to remain at the school because of bonuses. Retaining low-quality teachers would defeat the purpose of reducing turnover, so retaining teachers should not be assumed to be a fully positive metric. Even if low-performing teachers leave, then the school faces the problem of having to spend more money and effort on finding and training new teachers.

Recommendation and Implementation

As evidenced by some of the merit pay initiatives, simply throwing more money at a problem does not solve it. More money with no strings attached is not an effective intervention. Because of the lack of consistency, issues with using test scores to determine salary, and assuming that teachers are solely responsible for their student outcomes, teacher-level performance based pay does not seem to be the best option to reform traditional salary ladders. The more appealing financial incentive is offering bonuses to attract and retain teachers, especially in high-needs schools. A strategy with strong potential to enhance student learning is reducing the inequities in access to high quality educators. The bonuses mentioned in Option 2 can help keep effective teachers in hard-to-staff schools where they are typically more likely to leave. This will lead to the greatest overall increase in student achievement and work to narrow gaps between low-performing and high-performing schools. The policy should be focused at the elementary school level, since elementary school teachers spend the most time with their students and therefore have the highest chance of impact. Additionally, the Tennessee retention-bonus case demonstrated that financial incentives affected the behaviors and outcomes

of teachers in tested versus non-tested subjects differently. Hence, the most clear results will come from a policy aimed at elementary schools.

It may be difficult to compute the exact amounts necessary to provide an effective bonus, but further research and calculations based on school contexts can help policymakers reach a reasonable solution. The counterargument that one-time bonuses do not have long-lasting impacts is not fully supported by research. The initiatives have not been studied long enough yet to prove this point. Additionally, so long as schools continue to offer yearly retention bonuses, and do not just stop the program after one year, it is plausible that the positive effects of attracting and maintaining high quality teachers will persist. Finally, the strikes in Denver over ProComp could have been avoided with greater transparency. Because of this history, it is possible that attraction and retention bonus efforts will be met with initial backlash. However, to combat this, teachers need to be properly informed when they are eligible for a bonus and what the implications are. Yearly metrics should also be made more clear so that stakeholders understand how well the policy is working and any special cases where the policy is failing.

Evaluation

The main outcome that should change as a result of retention bonuses is teacher turnover rates. The combination of attracting and retaining high quality teachers should shift the overall composition of teachers in schools such that the average effectiveness of teachers increases. Then, because teacher quality and low turnover rates are linked to positive student outcomes, the policy should also lead to student growth and higher achievement, measured by test scores. The studied implementations have demonstrated higher retention rates and changes in schools' overall teacher quality after a year or two of these bonuses, but in order to be truly beneficial, the policy should also lead to longer-term effects beyond the initial year of attraction incentives.

Conclusion

While attempts at compensation reforms have had inconclusive impacts, the theories behind them provide reasonable suggestions for promoting teacher effectiveness and student achievement. The traditional payment schedule does not reward teachers for better performance, and it incentivizes teachers to move to the most privileged schools with the best resources. This causes inequities to persist, and there is a need to try to improve teacher quality in high needs schools. Because basing salaries on student test scores is unjust, a better alternative to single-salary schedules would be to introduce bonuses to bring effective teachers to high-poverty schools and retain them there for longer. Though there are only a few examples thus far of this approach, they demonstrate fairly positive results, and we can learn from their flaws to refine the policy and ultimately strengthen student outcomes.

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